

A bit of theory as far as relevant to the implementation of `class LagrangeMesh`

The necessary theory - for non-reduced axes - is found in [Ryssens et al, PHYSICAL REVIEW C 92, 064318 \(2015\)](#) section III.B Lagrange-mesh representation.

Note

The figures in this document are created by the tests. If the document complains `blablablah.png could not be found.`, please run the tests (`pytest tests` from folder `MOCCaPy`)

Non-reduced axes

(As from [Ryssens et al, PHYSICAL REVIEW C 92, 064318 \(2015\)](#) section III.B Lagrange-mesh representation)

Grid points

Using M evenly space grid point at distance dx , the boundaries of the box are $[-\frac{M}{2}dx, \frac{M}{2}dx]$. The width of the box is $L = Mdx$. We require $M = 2N$ to be even such that there are $N = \frac{M}{2}$ grid points on each side of the origin and the origin is avoided as a grid point.

For non-reduced axes (where the shift parameter σ is required be zero) the grid points are:
[eq 1]

$$\pm \frac{1}{2}dx, \pm \frac{3}{2}dx, \dots, \pm \frac{2N-3}{2}dx, \pm \frac{2N-1}{2}dx$$

or

[eq 2]

$$x_{\pm i} = \pm \left(\frac{1}{2} + i \right) dx = \pm \left(\frac{2i+1}{2} \right) dx$$
$$i = 0..N-1$$

in order:

[eq 3]

$$-\frac{2N-1}{2}dx, -\frac{2N-3}{2}dx, \dots, -\frac{3}{2}dx, -\frac{1}{2}dx, \frac{1}{2}dx, \frac{3}{2}dx, \dots, \frac{2N-3}{2}dx, \frac{2N-1}{2}dx$$

or

[eq 4]

$$x_i = -\frac{2N-1-2i}{2}dx = (\frac{1}{2} + i - N)dx, i = 0..(2N-1)$$

or

[eq 4.1]

$$x_i = -\frac{2N-1-2i}{2}dx = (\frac{1}{2} + i - N)dx, i = -(N-1)..(N-1)$$

Basis functions

The basis functions are plane waves:

[eq 5]

$$\phi_k(x) = \frac{1}{\sqrt{L}} \exp(\frac{2\pi i}{L} kx)$$

where i is the imaginary unit (not the index i), and

$$k = \pm\frac{1}{2}, \pm\frac{3}{2}, \dots, \pm\frac{2N-3}{2}, \pm\frac{2N-1}{2}$$

or, since kdx is a grid point, say, the i -th, x_i :

[eq 5.1]

$$\phi_{x_i}(x) = \frac{1}{\sqrt{L}} \exp(\frac{2\pi i}{L} \frac{x_i}{dx} x) = \frac{1}{\sqrt{L}} \exp(\frac{2\pi i}{Ldx} x_i x)$$

Note

Since $\exp(iz) = \cos z + i \sin z$ the real component of the 1D basis function is symmetric and the imaginary component is skew-symmetric. Consequently, we can use them for testing interpolation with Lagrange functions (see below) also on reduced grids.

When x is a grid point, e.g. $x = x_j$, we have

$$\begin{aligned} \phi_{x_i}(x_j) &= \frac{1}{\sqrt{L}} \exp(\frac{2\pi i}{Ldx} x_i x_j) = \frac{1}{\sqrt{L}} \exp(\frac{2\pi i}{2Ndx} (\frac{1}{2} + i - N)dx (\frac{1}{2} + j - N)dx)) \\ &= \frac{1}{\sqrt{L}} \exp(\frac{\pi i}{N} (\frac{1}{2} + i - N)(\frac{1}{2} + j - N)) \\ &= \frac{1}{\sqrt{L}} \exp(\frac{\pi i}{N} (\frac{1}{4} + \frac{1}{2}j - \frac{1}{2}N + i\frac{1}{2} + ij - iN - N\frac{1}{2} - Nj + N^2)) \\ &= \frac{1}{\sqrt{L}} \exp(\frac{\pi i}{N} (\frac{1}{4} + \frac{1}{2}(i+j) - N + ij - (i+j)N + N^2)) \end{aligned}$$

(this isn't very helpful.)

Interpolation

The Lagrange interpolation functions are:

[eq 6]

$$f_i(x) = \frac{1}{2N} \frac{\sin(\frac{\pi}{dx}(x - x_i dx))}{\sin(\frac{\pi}{dx} \frac{x - x_i dx}{2N})}$$

By construction, the Lagrange interpolation functions $f_i(x)$. have the property of being equal to 1 at the i -th mesh point, x_i , and 0 at all others:

[eq 7]

$$f_i(x_j) = \delta_{ij}$$

Note that when evaluating $f_i(x_i)$, both the numerator and the denominator are 0, and Python yields a `nan`. We need to invoke l'Hôpital's rule to obtain the result.

[eq 7.1]

$$\begin{aligned} & \lim_{x \rightarrow x_i} \frac{1}{2N} \frac{\sin(\frac{\pi}{dx}(x - x_i dx))}{\sin(\frac{\pi}{dx} \frac{x - x_i dx}{2N})} \\ &= \frac{1}{2N} \lim_{x \rightarrow x_i} \frac{\sin'(\frac{\pi}{dx}(x - x_i dx))}{\sin'(\frac{\pi}{dx} \frac{x - x_i dx}{2N})} \\ &= \frac{1}{2N} \lim_{x \rightarrow x_i} \frac{\frac{\pi}{dx} \cos(\frac{\pi}{dx}(x - x_i dx))}{\frac{\pi}{2N dx} \cos(\frac{\pi}{dx} \frac{x - x_i dx}{2N})} \\ &= \frac{2N}{2N} \lim_{x \rightarrow x_i} \frac{\cos(0)}{\cos(0)} = 1 \end{aligned}$$

In order to avoid testing for $\frac{0}{0}$ it suffices to add $\varepsilon \approx 1e - 9$ to x_i to avoid the `nan` and yield something close to 1.

The arguments of the two sine functions in eq 6 are the same, apart from a factor $1/2N$:

[eq 8]

$$f_i(x) = \frac{1}{2N} \frac{\sin(A_i(x))}{\sin(\frac{A_i(x)}{2N})}$$

$$A_i(x) = \frac{\pi}{dx}(x - x_i dx) = \pi(\frac{x}{dx} - x_i)$$

This can be exploited for efficiency in a computation.

An arbitrary function $h(x)$ taking the values $h_i = h(x_i)$ on the grid points can be interpolated on an arbitrary point $x \in [-Ndx, Ndx]$ as:

[eq 9]

$$h(x) = \sum_{i=0}^{2N-1} h(x_i) f_i(x)$$

This is essentially a dot product $\mathbf{f} \cdot \mathbf{h}$.

Derivatives

The 1st and 2nd order derivatives of the Lagrange interpolation functions are:
[eq 10.1]

$$D_{ji}^{(1)} = \left. \frac{df_i(x)}{dx} \right|_{x=x_j} = \begin{cases} (-1)^{i-j} \frac{\pi}{2Ndx} \frac{1}{\sin(\pi(i-j)/2N)}, & \text{for } i \neq j. \\ 0, & \text{for } i = j. \end{cases}$$

[eq 10.2]

$$D_{ji}^{(2)} = \left. \frac{d^2 f_i(x)}{dx^2} \right|_{x=x_j} = \begin{cases} (-1)^{i-j+1} 2 \left(\frac{\pi}{2Ndx} \right)^2 \frac{\cos(\pi(i-j)/2N)}{\sin^2(\pi(i-j)/2N)}, & \text{for } i \neq j. \\ -\frac{\pi^2}{3dx^2} \left(1 - \frac{1}{(2N)^2} \right), & \text{for } i = j. \end{cases}$$

By applying these to the expansion $\phi(x) = \sum_{i=0}^{2N-1} \phi(x_i) f_i(x)$ we obtain
[eq 11]

$$\frac{d\mathbf{h}}{dx} = \left. \frac{dh(x)}{dx} \right|_{x=x_j} = \sum_{i=0}^{2N-1} \left. \frac{df_i(x)}{dx} \right|_{x=x_j} h(x_i) = \sum_{i=0}^{2N-1} D_{ji}^{(1)} h(x_i) = \mathbf{D}^{(1)} \mathbf{h}$$

So, the column vector $\frac{d\mathbf{h}}{dx}$ of the derivatives of $\mathbf{h}$ at all grid points is found as a matrix product of $\mathbf{D}^{(1)}$ with the column vector of the values of $\mathbf{h}$ at all the gridpoints. As the 2nd derivative forms a matrix as well, we also have
[eq 12]

$$\frac{d^2 \mathbf{h}}{dx^2} = \mathbf{D}^{(2)} \mathbf{h}$$

Note, that single differentiation toggles the symmetry behavior of the function: if $h(x)$ is a symmetric function, $h'(x)$ is skew-symmetric, and *vice versa*. Consequentially, double differentiation toggles it twice, hence keeps the symmetry behaviour the same.

Reduced axes

Grid points

On reduced axes only the N grid points to the right of the origin are kept:
[eq 13]

$$\frac{1}{2}dx, \frac{3}{2}dx, \dots, \frac{2N-3}{2}dx, \frac{2N-1}{2}dx$$

or

[eq 14]

$$x_i = \left(i + \frac{1}{2}\right)dx$$

$$i = 0..N-1$$

(In the case of reduced axis a shift $\sigma < dx$ may be subtracted from each grid point.)

Interpolation

According to (eq 9) an arbitrary function $h(x)$ can be interpolated as:

$$h(x) = \sum_{i=0}^{2N-1} h(x_i) f_i(x)$$

If the x -axis is reduced, i.e. we require $h(x)$ to be symmetric ($h(-x) = h(x)$) or skew-symmetric ($h(-x) = -h(x)$), then the above equation becomes:

$$h(x) = \sum_{i=0}^{N-1} h(x_i) f_i(x) + \sum_{i=0}^{N-1} h(-x_i) f_{-i}(x)$$

where $f_{-i}(x)$ is the Lagrange interpolation function corresponding to the i -th grid point to the left of the origin

[eq 14.1]

$$f_{-i}(x) = \frac{1}{2N} \frac{\sin(\frac{\pi}{dx}(x - x_{-i}dx))}{\sin(\frac{\pi}{dx} \frac{x - x_{-i}dx}{2N})} = \frac{1}{2N} \frac{\sin(\frac{\pi}{dx}(x + (\frac{1}{2} + i)dx))}{\sin(\frac{\pi}{dx} \frac{x + (\frac{1}{2} + i)dx}{2N})}$$

which then becomes:

$$h(x) = \sum_{i=0}^{N-1} \begin{cases} h(x_i)(f_i(x) + f_{-i}(x)), & h \text{ is symmetric} \\ h(x_i)(f_i(x) - f_{-i}(x)), & h \text{ is skew-symmetric} \end{cases}$$

or

[eq 15]

$$h(x) = \sum_{i=0}^{N-1} h(x_i)(f_i(x) \pm f_{-i}) = \mathbf{h} \cdot \mathbf{f}_{\pm}$$

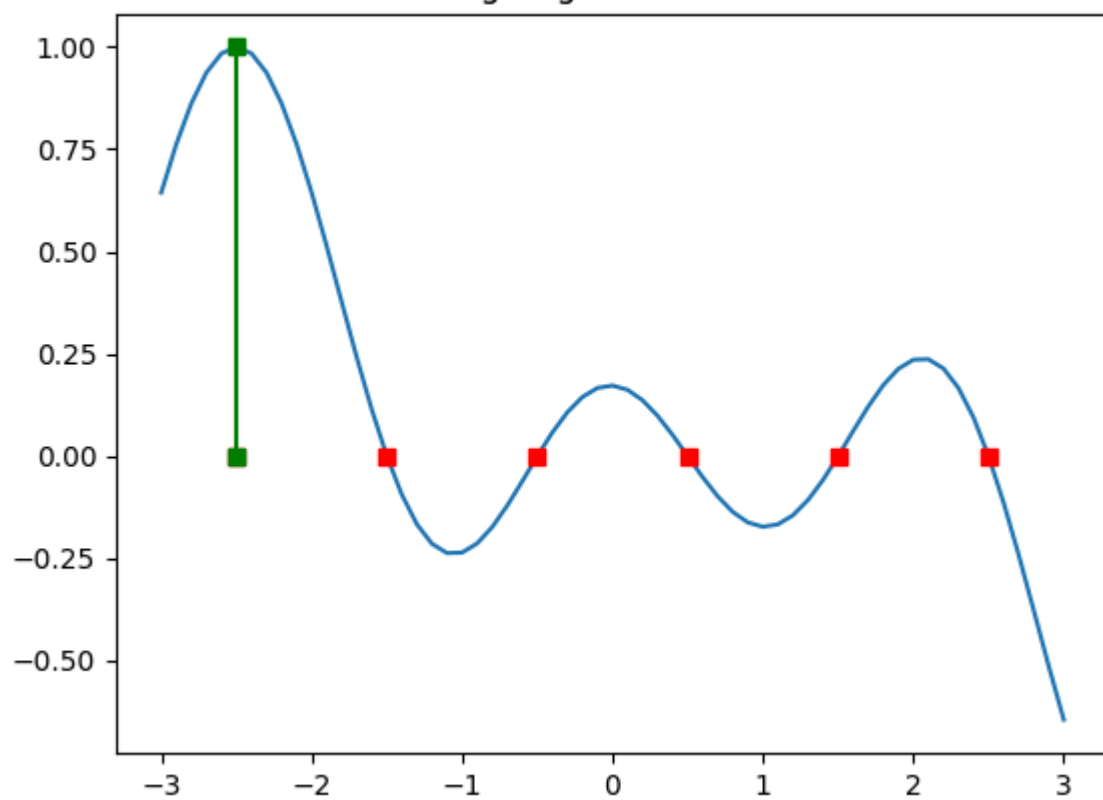
where the $\pm$ is $+$ for the symmetric case and $-$ for the skew-symmetric case, and $\mathbf{f}_{\pm} = \mathbf{f}_+ \pm \mathbf{f}_-$

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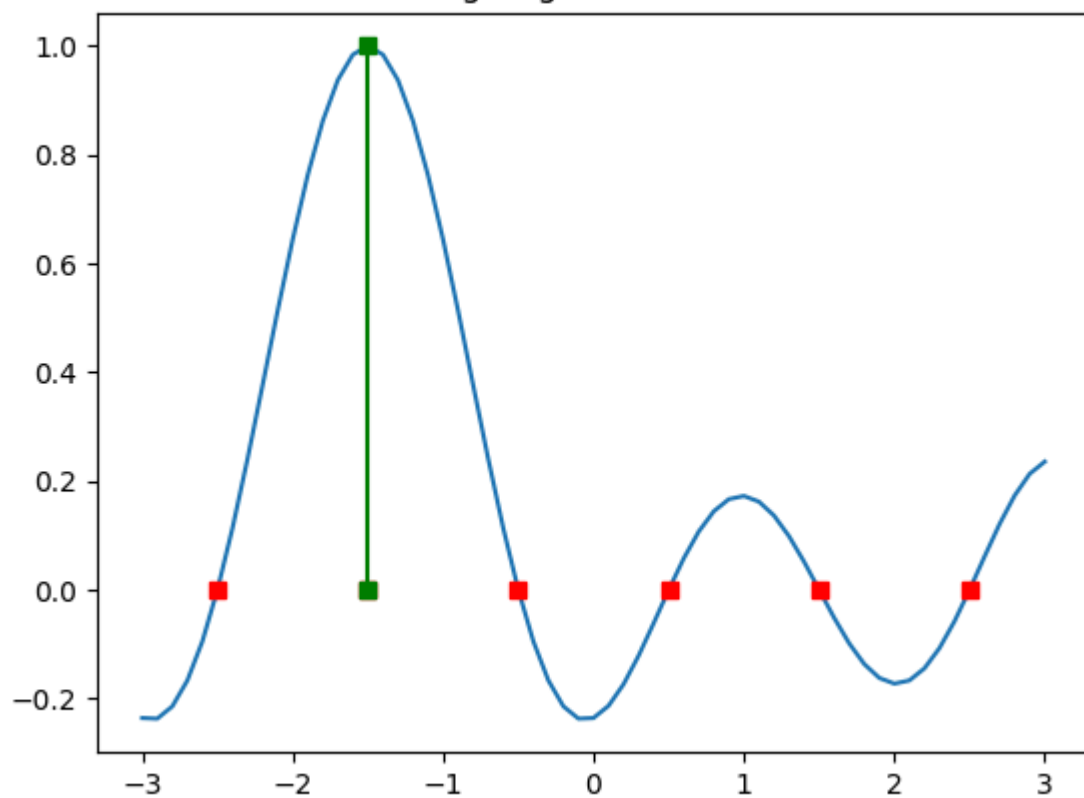
General remark on interpolation with Lagrange functions

On a 1D grid with 6 grid points at $-1.25, -.75, -.25, .25, .75, 1.25$ on the interval $[-1.5, 1.5]$, these are the 6 Lagrange functions:

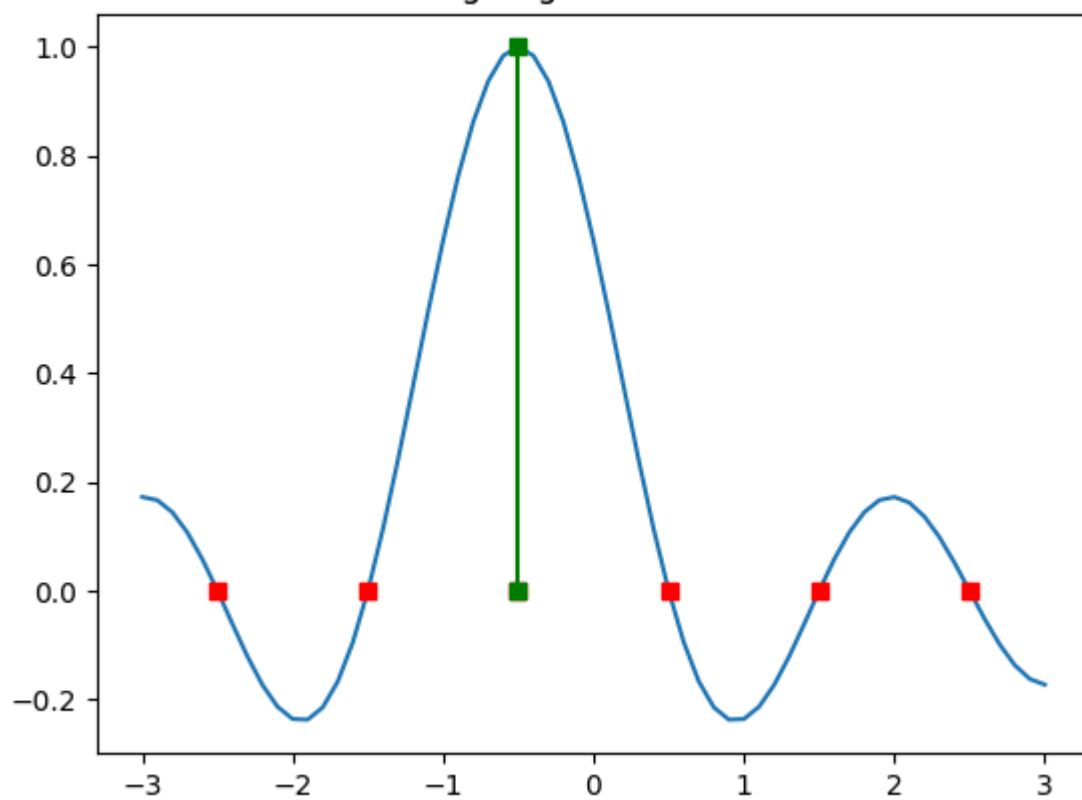
lagrange function 0



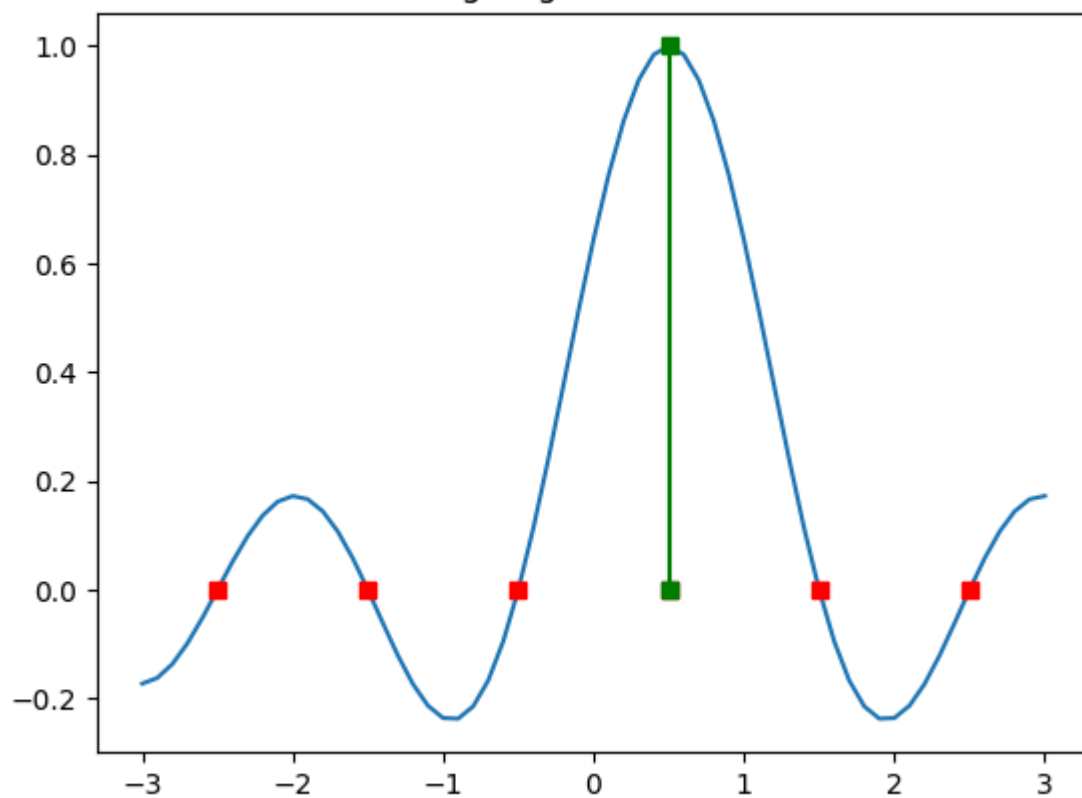
lagrange function 1



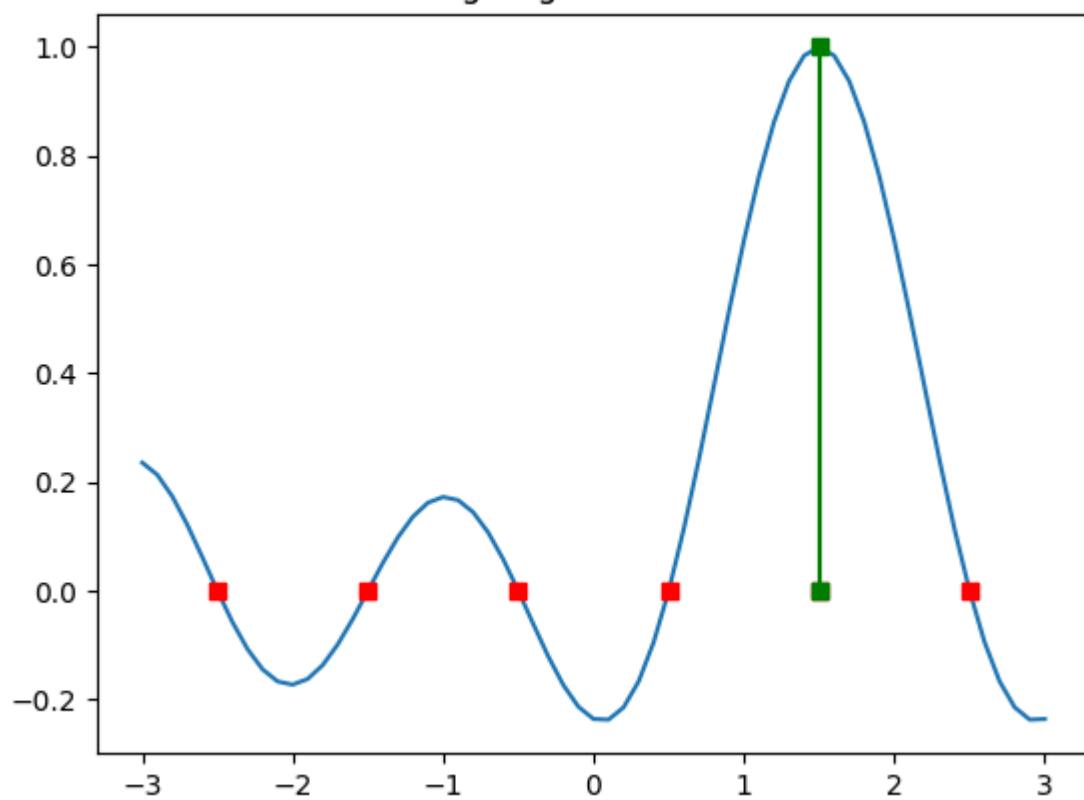
lagrange function 2



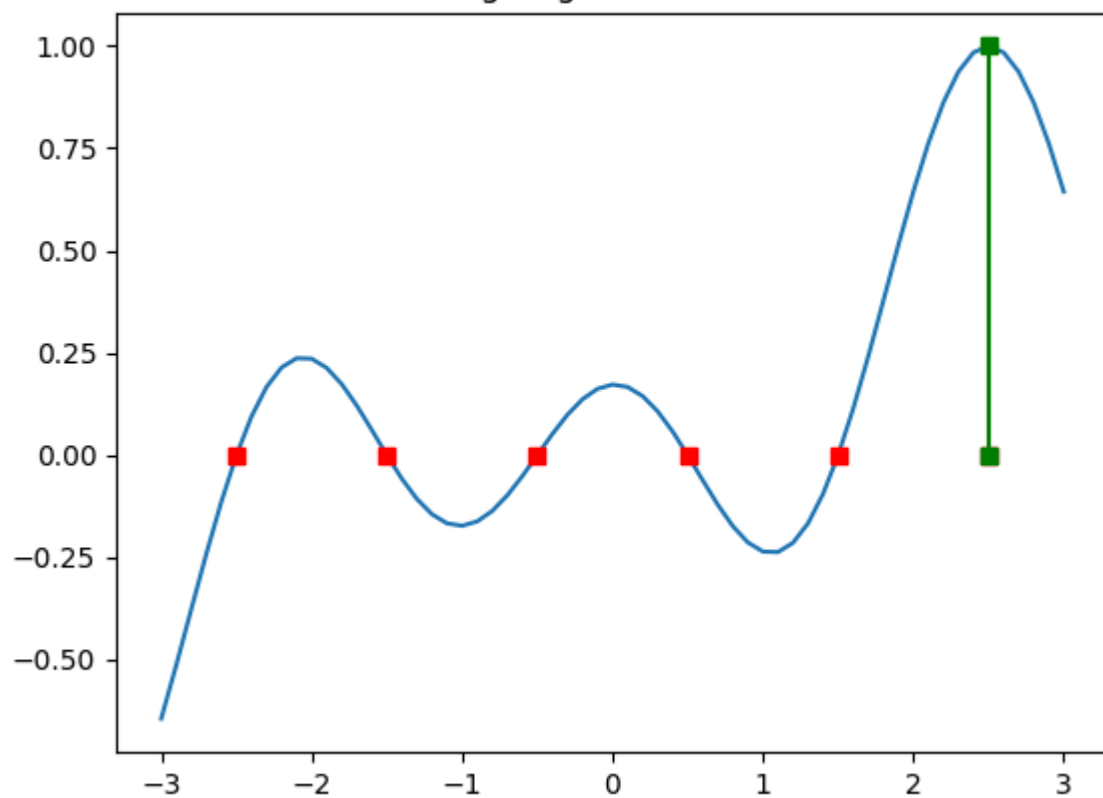
lagrange function 3

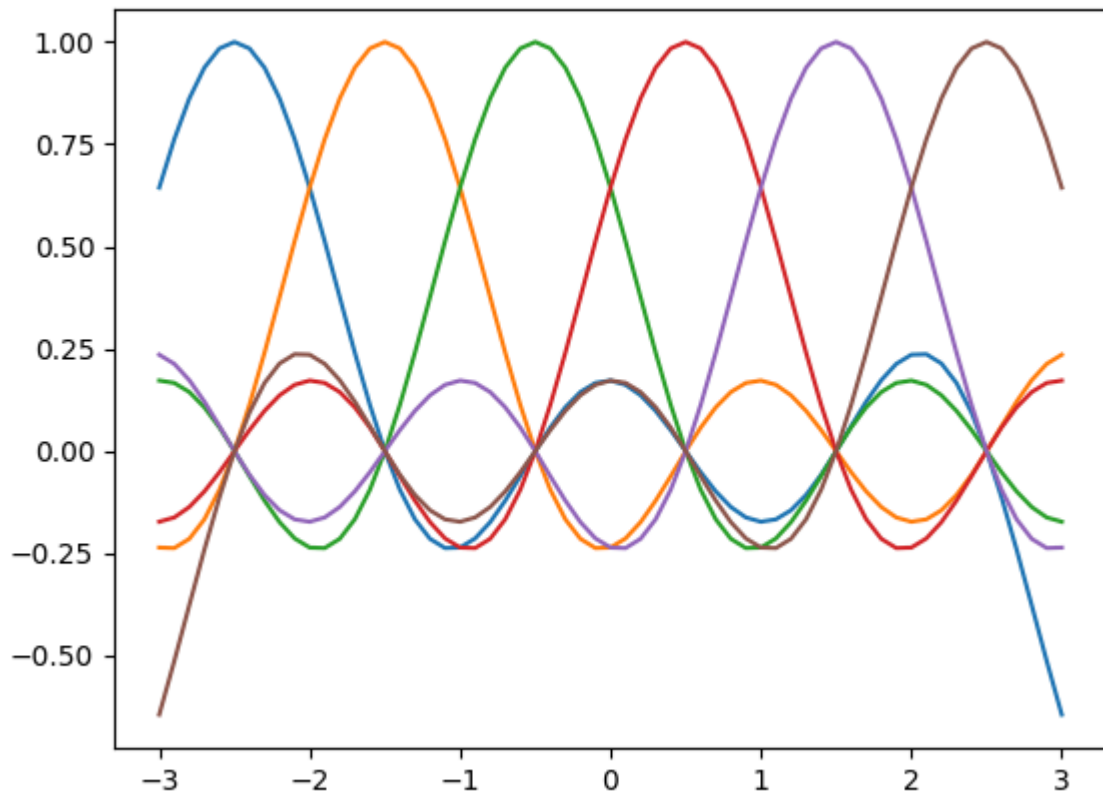


lagrange function 4



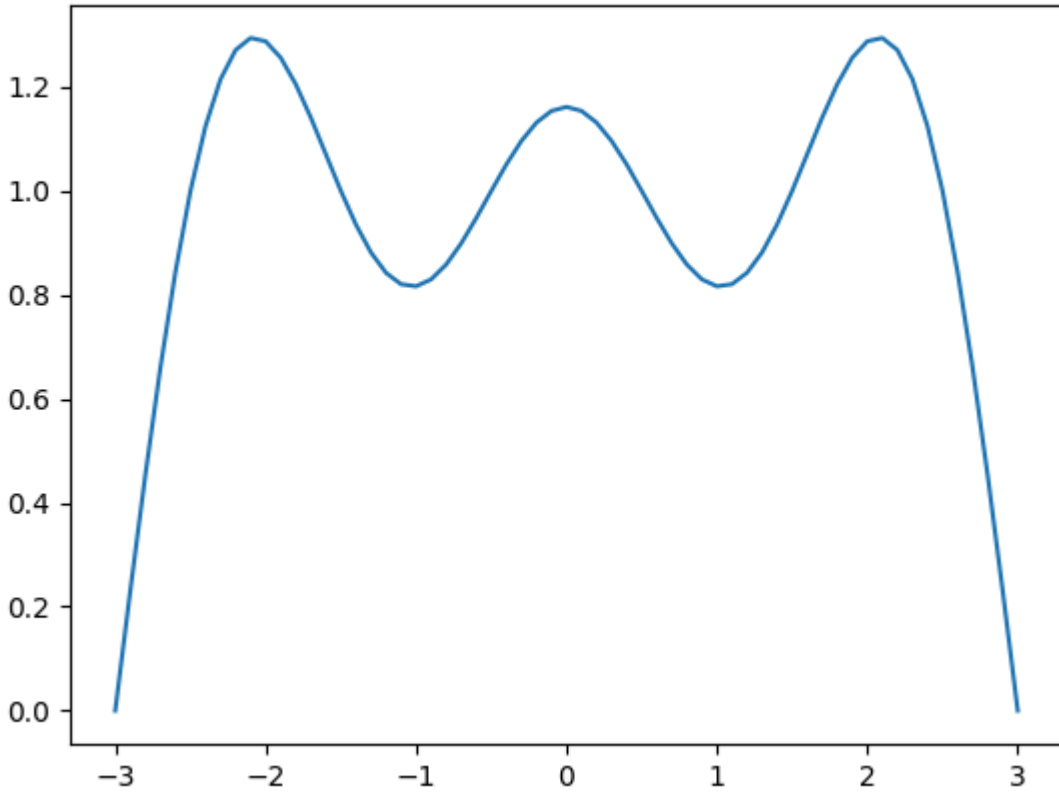
lagrange function 5





Clearly, each one yields 1 at one grid point and 0 at the others. Note also that they are antiperiodic: they reenter the box at the opposite edge, but **with a sign change**.

Consequently, a constant function cannot be interpolated because it is periodic. The sum of the 6 Lagrange functions is shown below. it is definitely not the constant function $f(x) = 1$.



According to the discussion in [github issue 52](#) periodic functions can be interpolated with Lagrange functions provided they vanish at the boundary of the interval.

Derivatives

The formula for the derivative of a function h expanded on a reduced grid is found easily by extending the column vector $\mathbf{h}$ (of length N) on the reduced grid as [eq 17]

$$\begin{bmatrix} \pm \mathbf{g} \\ \mathbf{h} \end{bmatrix}$$

with $\mathbf{g}$ containing the same elements as $\mathbf{h}$ but in reverse order, if the function h is symmetric, and its negatives in reverse order, if it skew-symmetric as if they were mirrored by a horizontal line between the two, and then forming the matrix product [eq 18]

$$\mathbf{D} \begin{bmatrix} \pm \mathbf{g} \\ \mathbf{h} \end{bmatrix} = \begin{bmatrix} \mathbf{D}^- \\ \mathbf{D}^+ \end{bmatrix} \begin{bmatrix} \pm \mathbf{g} \\ \mathbf{h} \end{bmatrix} = \begin{bmatrix} \mathbf{D}^- \begin{bmatrix} \pm \mathbf{g} \\ \mathbf{h} \end{bmatrix} \\ \mathbf{D}^+ \begin{bmatrix} \pm \mathbf{g} \\ \mathbf{h} \end{bmatrix} \end{bmatrix}$$

The upper half of $\mathbf{D}$, $\mathbf{D}^-$, produces the derivatives on the negative x -axis, and the lower half of $\mathbf{D}$, $\mathbf{D}^+$, produces the derivatives on the positive x -axis. The upper part and the lower part

are obviously related by symmetry: if the function $h(x)$ is symmetric, then the derivative is skew-symmetric and *vice versa*. Thus we are only interested in the lower part. Writing $\mathbf{D}^+$ as $[\mathbf{D}_{ll}|\mathbf{D}_{lr}]$ (ll stands for lower-left and lr for lower-right quadrant of $\mathbf{D}$) we find:
[eq 19]

$$\mathbf{D}^+ \begin{bmatrix} \pm \mathbf{g} \\ \mathbf{h} \end{bmatrix} = [\mathbf{D}_{ll}|\mathbf{D}_{lr}] \begin{bmatrix} \pm \mathbf{g} \\ \mathbf{h} \end{bmatrix} = [\mathbf{E}_{ll}|\mathbf{D}_{lr}] \begin{bmatrix} \pm \mathbf{h} \\ \mathbf{h} \end{bmatrix}$$

where $\mathbf{E}_{ll}$ is obtained from $\mathbf{D}_{ll}$ by reversing the order of the rows. Reversing the order of the rows of $\mathbf{D}_{ll}$ forces us to also reverse the order of the elements of $\pm \mathbf{g}$, which then becomes, by definition $\pm \mathbf{h}$. Finally, we obtain:
[eq 20]

$$\pm \mathbf{E}_{ll} \mathbf{h} + \mathbf{D}_{lr} \mathbf{h} = [\mathbf{D}_{lr} \pm \mathbf{E}_{ll}] \mathbf{h}$$

or

[eq 21]

$$\frac{d\mathbf{h}}{d\mathbf{x}} = [\mathbf{D}_{lr}^{(1)} \pm \mathbf{E}_{ll}^{(1)}] \mathbf{h}$$

where we have reintroduced the superscript ⁽¹⁾ to indicate the first order derivative. By the same reasoning, we obtain for the 2nd derivative:
[eq 22]

$$\frac{d^2\mathbf{h}}{d\mathbf{x}^2} = [\mathbf{D}_{lr}^{(2)} \pm \mathbf{E}_{ll}^{(2)}] \mathbf{h}$$

N-dimensional grids

The full Cartesian 3D representation of a function $h(\mathbf{r})$, where $\mathbf{r} = [x \ y \ z]$ (3D case), is then provided (for the 3D case) by
[eq 23]

$$h(\mathbf{r}) = \sum_{ijk} h_{ijk} f_i(x) f_j(y) f_k(z)$$

where the number of discretization points does not have to be the same in each direction.

Note that f_i , f_j and f_k are generally different objects, even if accidentally the indices i , j and k are identical, as they pertain, resp., to the x -axis, the y -axis and the z -axis, and the grid spacing is not necessarily the same.

Derivatives

In this case, the derivative matrices $\mathbf{D}^{(1)}$ and $\mathbf{D}^{(2)}$ have to be set up separately for each direction, taking into account whether the axis is reduced or not.

$$\left. \frac{d\Phi}{dx} \right|_{ijk} = \sum_l D_{il}^1 \Phi_{ljk}$$

$$\left. \frac{d\Phi}{dy} \right|_{ijk} = \sum_l D_{il}^1 \Phi_{ilk}$$

$$\left. \frac{d\Phi}{dz} \right|_{ijk} = \sum_l D_{il}^1 \Phi_{ijl}$$

The summation is over the index of Φ_{ijk} that corresponds to the axis wrt which the derivative is taken: the derivative wrt $x/y/z$ sums over the 1st/2nd/3rd index. The same holds for 2nd, 3rd, 4th order derivatives ($\frac{d^n}{du^n}$, $u = x, y, z$, $n \geq 0$).

Basis functions

The basis functions are plane wave products of the different axes:

[eq 24]

$$\begin{aligned} \Phi_{klm}(x, y, z) &= \phi_k(x) \phi_l(y) \phi_m(z) = \frac{1}{\sqrt{L_x}} \frac{1}{\sqrt{L_y}} \frac{1}{\sqrt{L_z}} \exp\left(\frac{2\pi i}{L_x} kx\right) \exp\left(\frac{2\pi i}{L_y} ly\right) \exp\left(\frac{2\pi i}{L_z} mz\right) \\ &= \frac{1}{\sqrt{L_x L_y L_z}} \exp\left(2\pi i \left(\frac{kx}{L_x} + \frac{ly}{L_y} + \frac{mz}{L_z}\right)\right) \\ k &= \pm \frac{1}{2}, \pm \frac{3}{2}, \dots, \pm \frac{N_x - 1}{2} \\ l &= \pm \frac{1}{2}, \pm \frac{3}{2}, \dots, \pm \frac{N_y - 1}{2} \\ m &= \pm \frac{1}{2}, \pm \frac{3}{2}, \dots, \pm \frac{N_z - 1}{2} \\ &= \frac{1}{\sqrt{L_x L_y L_z}} \exp(2\pi i (\mathbf{k} \cdot \mathbf{r})) \\ \mathbf{k} &= \begin{bmatrix} \frac{k}{L_x} & \frac{l}{L_y} & \frac{m}{L_z} \end{bmatrix} \\ \mathbf{x} &= [x \quad y \quad z] \end{aligned}$$

Note that ϕ_k , ϕ_l and ϕ_m are generally different objects, even if accidentally the indices k , l and m are identical, as they pertain, resp., to the x -axis, the y -axis and the z -axis.

Alternatively, in the spirit of eq 5.1:

[eq 24.1]

$$\mathbf{k} = \begin{bmatrix} \frac{x_i}{L_x dx} & \frac{y_j}{L_y dy} & \frac{z_k}{L_z dz} \end{bmatrix}$$

where the x_i , y_j , z_k are grid coordinates in the x , y and z directions.

Note

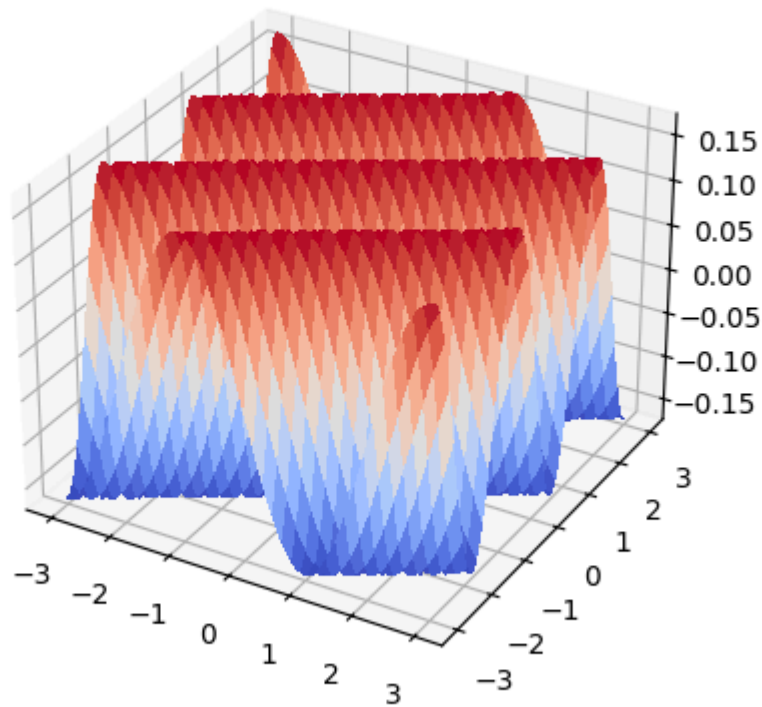
Contrary to the 1D case, it is generally not true for the 2D and 3D cases that the basis functions are symmetric or skew-symmetric.

E.g. changing the sign of the x -coordinate in $\exp(2\pi i(\mathbf{k} \cdot \mathbf{r})) = \exp(2\pi i(\mathbf{k} \cdot [-x, y, z]))$ yields

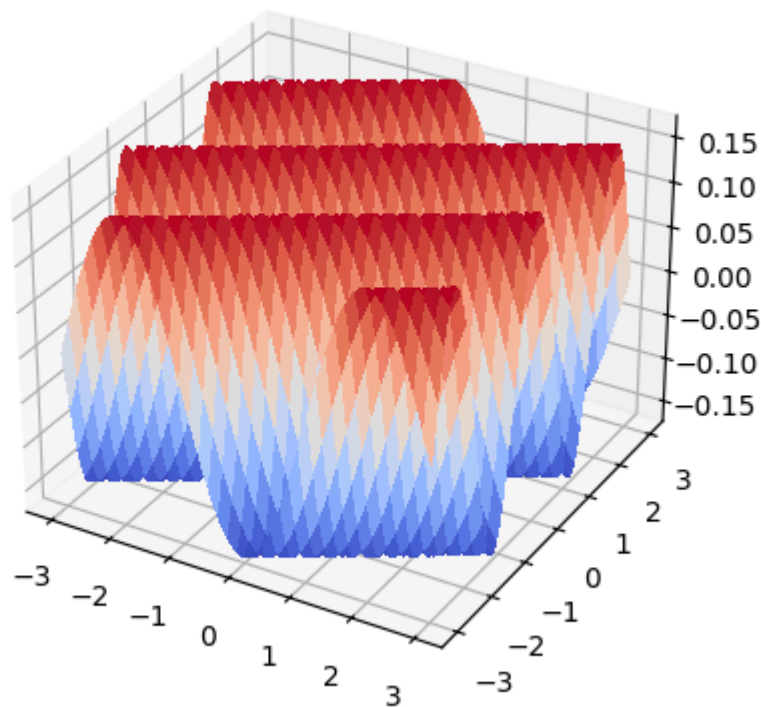
$$\cos(-k_x x + k_y y + k_z z) + i \sin(-k_x x + k_y y + k_z z)$$

and $\cos(-k_x x + k_y y + k_z z) = \cos(k_x x + k_y y + k_z z)$ only if $k_y y + k_z z$ is a multiple of 2π , which is generally not true. This can also be seen in the figures below. The plane waves are periodic only in the direction of the wave vector.

i=1, x_i=-1.5, y_j=2.5 real



i=1, x_i=-1.5, y_j=2.5 imag



Interpolation

As described by eq 23 Interpolating a scalar quantity h at a single point requires a sum over all grid points which may be costly (speaking of working interactively). If h needs to be interpolated on a large number of points, p ,

$$\mathbf{r} = \begin{bmatrix} x_0 & y_0 & z_0 \\ x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ \vdots & \vdots & \vdots \\ x_{p-1} & y_{p-1} & z_{p-1} \end{bmatrix}$$

it will be advantageous to move the loop over the points $0..p-1$ inside the loop over ijk product $h_{ijk}f_i(x)f_j(y)f_k(z)$.

In case h is not a scalar quantity but a tensor it is advantageous to move the loop over the components between the loop over the p interpolation points and the loop over ijk :

```
for all grid points ijk:
    for all components h of H
        for all interpolation points 0..p
            accumulate h_ijk * f_i(x_p) * f_j(y_p) * f_k(z_p) over ijk
```

The inner loop can be evaluated as a function over a numpy array:

```
for all grid points ijk:
    for all components h of H
        h(r) += h_ijk * f_i(r[:,0]) * f_j(r[:,1]) * f_k(r[:,2])
```

Furthermore, in case e.g. the x axis is reduced, according to eq 15 one must replace $f_i(x)$ by $(f_i(x) \pm f_{-i}(x))$, where the sign is $+$ if h is symmetric w.r.t. x and $-$ if h is skew-symmetric w.r.t. x .